WORKED SOLUTION

STEP I 1991, Question 3

Pure • Complex Geometric Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q3

The key idea

In the Argand diagram, multiplying by $re^{i\phi}$ scales a vector by r and turns it anticlockwise through ϕ. Each segment of the path is the previous one scaled and turned by the same amount, so each segment vector is the previous one times a fixed complex number. The segments then form a geometric progression, and adding them up gives the position of every vertex.

Solution

The ratio of successive segments

The vector from P_n to P_{n+1} is represented by $z_{n+1} - z_n$, and the vector from P_{n-1} to P_n by $z_n - z_{n-1}$. For the quotient of two complex numbers, moduli divide and arguments subtract:

$$\left|\frac{z_{n+1} - z_n}{z_n - z_{n-1}}\right| = \frac{P_nP_{n+1}}{P_{n-1}P_n} = d, \qquad \arg\frac{z_{n+1} - z_n}{z_n - z_{n-1}} = \arg(z_{n+1} - z_n) - \arg(z_n - z_{n-1}) = \theta,$$

because the second segment is d times as long as the first, and the anticlockwise angle from the direction of one directed segment to the direction of the next is θ. (The argument is only defined up to multiples of 2π, which doesn't affect the complex number.) Hence, for every $n \geq 2$,

$$\frac{z_{n+1} - z_n}{z_n - z_{n-1}} = d\,e^{i\theta} = d(\cos\theta + i\sin\theta).$$

A formula for z_{n+1}

Write $w = de^{i\theta}$. The first segment is $z_2 - z_1 = 1$, and each later segment is w times the one before, so by induction

$$z_{k+1} - z_k = w^{k-1} \qquad (k \geq 1).$$

Adding these for $k = 1, 2, \ldots, n$, the left-hand side telescopes to $z_{n+1} - z_1 = z_{n+1}$:

$$z_{n+1} = 1 + w + w^2 + \cdots + w^{n-1} = \frac{w^n - 1}{w - 1} = \frac{d^n \mathrm{e}^{\mathrm{i}n\theta} - 1}{d\mathrm{e}^{\mathrm{i}\theta} - 1}.$$

Summing the geometric series needs $w \neq 1$, and this is where $d \neq 1$ is used, since $|w| = d$.

The case $\theta = \frac{1}{3}\pi$, $d = 2$

Now $w = 2\mathrm{e}^{\mathrm{i}\pi/3} = 1 + \mathrm{i}\sqrt{3}$, so the denominator is very simple: $w - 1 = \mathrm{i}\sqrt{3}$. Since $\frac{1}{\mathrm{i}} = -\mathrm{i}$,

$$z_{n+1} = \frac{w^n - 1}{\mathrm{i}\sqrt{3}} = -\frac{\mathrm{i}}{\sqrt{3}}\left(2^n \cos\tfrac{n\pi}{3} - 1 + \mathrm{i}\, 2^n \sin\tfrac{n\pi}{3}\right),$$

so that

$$\operatorname{Re} z_{n+1} = \frac{2^n \sin\frac{n\pi}{3}}{\sqrt{3}}, \qquad \operatorname{Im} z_{n+1} = \frac{1 - 2^n \cos\frac{n\pi}{3}}{\sqrt{3}}.$$

Ask yourself How does the sign of $\operatorname{Im} z_{n+1}$ tell us that the path crosses the real axis?

The path is joined up: it is a continuous chain of segments. If one vertex lies strictly below the real axis and a later vertex lies strictly above it (or the other way round), then the stretch of path between them must cross the real axis. So we look for vertices on opposite sides, again and again.

The value of $\cos\frac{n\pi}{3}$ repeats with period 6. Take any integer $k \geq 1$.

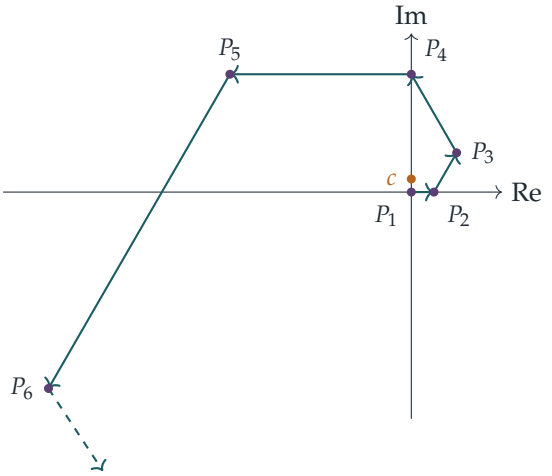
- For $n = 6k$, $\cos\frac{n\pi}{3} = \cos 2k\pi = 1$, so $\operatorname{Im} z_{6k+1} = \dfrac{1 - 2^{6k}}{\sqrt{3}} < 0$, since $2^{6k} > 1$.
- For $n = 6k + 3$, $\cos\frac{n\pi}{3} = \cos(2k\pi + \pi) = -1$, so $\operatorname{Im} z_{6k+4} = \dfrac{1 + 2^{6k+3}}{\sqrt{3}} > 0$.

Thus P_{6k+1} is below the real axis and P_{6k+4} is above it, for every $k \geq 1$. So the part of the path from P_{6k+1} to P_{6k+4} crosses the real axis, going upwards, for each $k = 1, 2, 3, \ldots$. These are different parts of the path, one for each k, so the path crosses the real axis infinitely often. (By the same argument it also crosses downwards between P_{6k+4} and P_{6k+7}.) This completes the proof.

In fact the vertices go round in blocks of three: the sign pattern of $\operatorname{Im} z_1, \operatorname{Im} z_2, \ldots$ is $0, 0, +, +, +, -, -, -, +, +, +, \ldots$. There is a neat geometric reason. Writing

$$z_{n+1} = \frac{\mathrm{i}}{\sqrt{3}} - \frac{\mathrm{i}}{\sqrt{3}}\, w^n,$$

we see that P_{n+1} is at distance $2^n/\sqrt{3}$ from the fixed point $c = \mathrm{i}/\sqrt{3}$, and each step turns it by 60° about c. The vertices lie on a spiral round c that doubles in size every step, and c sits just above the real axis, so the spiral keeps sweeping across the axis.



The figure shows $P_1, \ldots, P_6$; the next vertex is $P_7 = -21\sqrt{3}\,\mathrm{i}$, far below. The path first crosses the real axis on P_5P_6, then stays below it until it crosses upwards on P_8P_9, from $P_8 = 64 - 21\sqrt{3}\,\mathrm{i}$ to $P_9 = 128 + 43\sqrt{3}\,\mathrm{i}$.

Checks

- Directly from the geometry: $z_3 = 1 + w = 2 + \sqrt{3}\,\mathrm{i}$, and the formula gives $\operatorname{Im} z_3 = (1 - 4\cos\frac{2\pi}{3})/\sqrt{3} = 3/\sqrt{3} = \sqrt{3}$. Similarly, adding $w^2 = -2 + 2\sqrt{3}\,\mathrm{i}$ and $w^3 = -8$ gives $z_5 = -8 + 3\sqrt{3}\,\mathrm{i}$, and the formulas with $n = 4$ give $\operatorname{Re} z_5 = 16\sin\frac{4\pi}{3}/\sqrt{3} = -8$ and $\operatorname{Im} z_5 = (1+8)/\sqrt{3} = 3\sqrt{3}$.
- With $d = 1$ the vertices would lie on a regular polygon and the path would close up; the geometric-series formula fails exactly then, as it should.
- We built the path numerically from lengths and turning angles (random d, θ and starting point) and confirmed the ratio $d\mathrm{e}^{\mathrm{i}\theta}$, the closed form for z_{n+1}, and, for $\theta = \frac{1}{3}\pi$, $d = 2$, the sign pattern above: there are 13 sign changes of $\operatorname{Im} z$ among $z_1, \ldots, z_{41}$.

Common slips

- Writing the ratio as $\frac{1}{d}\mathrm{e}^{-\mathrm{i}\theta}$: the *later* segment is on top, so it is the one that is d times longer and turned through $+\theta$.
- An off-by-one error in the sum: there are n segments from P_1 to P_{n+1}, so the series runs up to w^{n-1}.
- Claiming a crossing from one vertex on the axis: you need vertices *strictly* on opposite sides (here z_1 and z_2 lie on the axis, so start from $k \geq 1$).

Beyond STEP

The rule $z_{n+1} - z_n = w(z_n - z_{n-1})$ is a linear recurrence, and the map $z \mapsto c + w(z - c)$ that carries each vertex to the next is a *spiral similarity* about its fixed point c: a rotation combined with an enlargement. Such maps are the simplest examples of Möbius transformations, and iterating them produces the equiangular (logarithmic) spirals seen in shells. Needham's *Visual Complex Analysis* develops this geometric way of thinking about complex multiplication beautifully.

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